

Week 8 Discussion: Options Basics & No-Arbitrage Properties

Main focus (25min): (i) intrinsic vs. time value; (ii) no-arbitrage bounds; (iii) put-call parity and early exercise intuition.

Module A) Intrinsic Value (IV) & Time Value (TV)

Option Price Components: $\text{Option Price}_t = \text{IV}_t + \text{TV}_t$

- **Intrinsic Value (IV):** The payoff if you were forced to exercise *right now*.

$$\text{IV}_t^{\text{call}} = (S_t - K)^+, \quad \text{IV}_t^{\text{put}} = (K - S_t)^+.$$

- **Time Value (TV):** The “hope” or “insurance” premium. It is the extra amount the market charges for the possibility that the stock will move favorably before expiration. $\text{TV}_t \geq 0$, and $\text{TV}_T = 0$ at maturity.

Intuition: Exercising an option early captures only the IV but instantly destroys the remaining TV.

Practice A (Textbook 10.11): Intrinsic value as a lower bound (American)

Explain why an American option is always worth at least as much as its intrinsic value.

Module B) Arbitrage-Free Bounds (Non-dividend stock)

- **Upper Bounds:** $C_t \leq S_t$ and $P_t \leq Ke^{-r(T-t)}$.

Intuition: A right to buy the stock can't cost more than the stock itself. A right to sell for K can't be worth more than the present value of the maximum possible payoff (K).

- **Lower Bounds:**

$$C_t \geq (S_t - Ke^{-r(T-t)})^+ \quad \text{and} \quad P_t \geq (Ke^{-r(T-t)} - S_t)^+$$

Intuition (Call Lower Bound): Imagine two portfolios:

Portfolio A: Buy Call + Save PV of K in the bank. (Worth $C_t + Ke^{-r(T-t)}$)

Portfolio B: Buy the Stock. (Worth S_t)

At maturity, Portfolio A *always* does at least as well as Portfolio B. Therefore, it must cost at least as much today.

Practice B (Textbook 11.1): Lower bound for a call

What is a lower bound for the price of a 4-month call option on a non-dividend-paying stock when $S_0 = \$28$, $K = \$25$, and $r = 8\%$ per annum?

Module C) Put–Call Parity & Early Exercise (Non-dividend stock)**Put–Call Parity (European):**

$$C_t(K, T) - P_t(K, T) = S_t - Ke^{-r(T-t)}.$$

Intuition (Synthetic Stock): Buying a call and selling a put exactly replicates the payoff of buying the stock using borrowed money (Forward contract).

American Call (no dividends): $C_t = C_t$. It is *never* optimal to exercise early. Why?

1. **Time Value of Money (TVM):** Why pay the strike price K today when you can keep the cash in the bank, earn interest, and pay K at maturity?
2. **Killing the Insurance:** Exercising kills the remaining Time Value (TV). If you want out of the position, you should *sell* the call in the market to capture (IV + TV) rather than exercising to only get IV.

Practice C1 (Textbook 11.6): Put–call parity computation

The price of a non-dividend-paying stock is \$19 and the price of a 3-month European call option on the stock with a strike price of \$20 is \$1. The risk-free rate is 4% per annum. What is the price of a 3-month European put option with a strike price of \$20?

Practice C2 (Textbook 11.3): Early exercise (conceptual)

Give **two** reasons why the early exercise of an American call option on a non-dividend-paying stock is not optimal. The first reason should involve the **time value of money**. The second should apply even if **interest rates are zero**.